RESEARCH

UAV 3D Path Planning Based on Integrated Particle Swarm Optimization and Artificial Potential Field Method

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Abstract

Safe, efficient, and feasible path planning for unmanned aerial vehicle (UAV) in complex three-dimensional environments remains a critical challenge in intelligent navigation. Although the traditional artificial potential field (APF) method offers high computational efficiency, it is prone to local minima and target inaccessibility in complex environments. To overcome these limitations, this paper proposes a novel 3D path planning approach that integrates an APF with particle swarm optimization (PSO). The method introduces a Gaussian repulsive function to improve obstacle boundary perception and ensure smooth potential field transitions. Combined with a PSO-based optimization of trajectory points, it effectively enhances the algorithm's ability to escape local minima and improves global path accessibility. To validate the effectiveness of the proposed method, a comprehensive evaluation framework was developed, incorporating key performance metrics such as total path length, average safety distance, and minimum safety distance. Benchmark comparisons were conducted against several advanced path planning algorithms in static, dynamic, and more complex obstacle environments. In addition, a path executability verification framework was introduced, employing multiple flight controllers to perform trajectory tracking simulations and systematically assess control feasibility and adaptability in complex dynamic scenarios. Experimental results show that the proposed method exhibits clear advantages in both path planning quality and practical executability, particularly demonstrating strong adaptability and robustness in complex environments.

Keywords Unmanned aerial vehicle · Path planning · Artificial potential field method · Particle swarm optimization algorithm · Gaussian repulsive function · Path executability verification

1 Introduction

In recent years, unmanned aerial vehicle (UAV) technology [1] has developed rapidly and found wide application in diverse critical domains, including military reconnaissance, disaster response, environmental monitoring, agricultural and forestry protection, as well as logistics and transportation [2]. This trend highlights UAVs' vast development prospects and enormous application potential. As a new generation of intelligent aerial platforms, UAVs offer several advantages such as flexible deployment, low cost, and ease of operation. These features are particularly valuable in complex terrains or high-risk operational environments, where UAVs can effectively substitute for human labor, thereby significantly enhancing both operational efficiency and safety [3]. With the growing complexity of application scenarios and the increasing demands for task execution accuracy, autonomous flight capability has emerged as a central focus in UAV technology development. Among the enabling technologies, path planning has garnered widespread attention from researchers, as it plays a critical role in ensuring the safe and efficient completion of UAV missions [4].

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Path planning technology aims to generate an obstacle-avoiding, safe, smooth, and efficient flight trajectory for unmanned aerial vehicles from a starting point to a target destination [5]. This task becomes particularly challenging in three-dimensional (3D) space due to the increased complexity of environmental obstacle distribution and stricter path feasibility constraints. Effective 3D path planning must not only avoid both static and dynamic obstacles but also optimize multiple performance metrics, including path length, flight time, and trajectory smoothness [6]. These requirements place greater demands on traditional planning methods regarding algorithmic efficiency, robustness, and adaptability.

The Artificial Potential Field (APF) method has been widely applied in robot and UAV path planning due to its simplicity, high computational efficiency, and suitability for dynamic environments. By simulating the interactions between attractive and repulsive forces, APF guides UAVs toward the target while steering them away from obstacles [7]. However, the classical APF approach exhibits significant limitations in practical applications, such as susceptibility to local minima, path oscillations, target inaccessibility, and poor adaptability to complex environments. These issues become particularly pronounced in densely obstructed scenarios or when the target is fully surrounded, leading to substantial degradation in path planning performance [8].

To overcome these limitations, researchers have increasingly focused on integrating APF with swarm intelligence optimization algorithms. Among these, Particle Swarm Optimization (PSO) [9], a representative global optimization technique, has shown strong capabilities in exploring the search space by simulating swarm cooperation. Its effective global search ability helps mitigate the local minima problem inherent in APF, making it a promising candidate for path planning tasks. However, PSO itself suffers from challenges such as premature convergence, limited local search capabilities, and sensitivity to parameter settings, which can compromise the stability and accuracy of optimization results.

To address these challenges, this study introduces a novel three-dimensional UAV path planning method, termed FPAPF (Fusion of Particle Swarm Optimization and Artificial Potential Field). The approach leverages the directional guidance of APF for rapid initial path generation, employing an attractive potential combined with a modified Gaussian repulsive function. Building on this, PSO is introduced in a guided manner to locally refine the trajectory points generated by APF, enhancing overall path quality and robustness. This fusion effectively combines the computational efficiency and directional guidance of APF with the exploratory optimization capability of PSO, resulting in improved path feasibility and adaptability in complex environments.

Moreover, to ensure the practical applicability of the planned paths, a comprehensive evaluation framework is developed, incorporating key performance metrics such as path length and safety distance. This framework supports assessment in both static and dynamic obstacle environments, providing a holistic view of path quality under varied conditions. Furthermore, a path executability verification mechanism is proposed, extending evaluation beyond the planning stage into the execution phase through controller-based simulation experiments and trajectory tracking analyses.

The main contributions of this study are summarized as follows:

(1) Proposes an improved Gaussian-based repulsive function that enhances obstacle perception and addresses discontinuity issues near environmental boundaries in traditional APF.
(2) Establishes a guided optimization mechanism by integrating PSO with APF, enabling dynamic path adjustment to escape local minima and overcome target inaccessibility, thereby improving planning robustness and success rates.
(3) Designs a multi-metric evaluation framework incorporating path length, average safety distance, and minimum safety distance, which systematically assesses algorithms across static, dynamic, and complex environments, demonstrating the superior efficiency, safety, and the adaptability of the proposed method.
(4) Develops a unified controller evaluation framework to verify path executability under consistent inputs. Through disturbance injection and non-smooth trajectory testing, it quantitatively evaluates the robustness and tracking performance of the control strategies in complex scenarios.

The structure of this paper is as follows: Sect. 2 reviews related work; Sect. 3 describes the principles and implementation of the proposed hybrid algorithm in detail; Sect. 4 presents the simulation setup and analyzes the experimental results; and the final section summarizes the paper and outlines future research directions.

2 Related Work

The objective of unmanned aerial vehicle (UAV) path planning is to determine a feasible route from a starting point to a destination within a known or partially known environment. This route must ensure obstacle avoidance while optimizing factors such as path length, smoothness, and flight safety. With advances in algorithms and computational technologies, path planning methods can be broadly categorized into graph-based search algorithms, sampling-based methods and hybrid strategies.

Graph-based algorithms such as A* and Dijkstra are widely used for their stability and global optimality [10, 11]. For A*, this guarantee relies on heuristics that are both admissible and consistent; if only admissibility holds, optimality and reduced expansions may fail under extreme counterexamples [12]. This dependence underscores the challenge of heuristic design in practice. 3D-constrained Hybrid A* extends the 2D variant to three-dimensional alignment planning by embedding earthwork, land-take, and path-length costs into the objective. Search is accelerated through discrete cross-sectional integration and geometric pruning, but the estimates lack a uniform error bound and are sensitive to grid resolution. Excessive pruning may discard valid branches, while scalability in large domains with multiple constraints remains limited [13]. Common acceleration strategies modify the evaluation function. for example, incorporating RBF-based dynamic weights and heuristic bonuses into f=g+h can reduce expansions and improve speed, but typically breaks heuristic consistency, voiding optimality guarantees. In elongated obstacle settings, such modifications may also induce repeated open-list revisits and increase runtime [14]. Despite various acceleration strategies, these algorithms remain sensitive to discretization and struggle with scalability in complex 3D environments.

Sampling-based methods, such as Rapidly-exploring Random Tree (RRT) [15], scale well in high-dimensional spaces but often generate non-smooth, hard-to-control paths. However, they face several intrinsic limitations: (i) in narrow or high-dimensional spaces, the feasible region occupies only a small fraction of the volume, making effective samples rare and convergence slow. Informed RRT*'s ellipsoidal sampler may degenerate to near-global sampling in corridor-like environments, while achieving satisfactory solutions often requires large sample sizes and memory [16]. (ii) Most approaches ensure only geometric collision avoidance, without enforcing dynamics or continuous-time safety margins. When exact steering functions are unavailable and discrete motion primitives are used, both optimality and safety guarantees weaken. For example, articulated vehicles in underground mines cannot meet turning-radius constraints with vanilla RRT*, requiring additional rules such as adaptive step size and turn-angle limits [17]. (iii) Evaluation practices remain limited, often focusing on path length, time, or iterations under fixed hyperparameters. Broader analyses of smoothness, clearance, execution cost, replanning robustness, and hyperparameter sensitivity are still insufficient [18]. Although RRT variants scale well in theory, their practical utility is hindered by inefficiency in constrained domains and insufficient consideration of dynamics and robustness.

Despite the progress of graph-based and sampling-based algorithms, their limitations in scalability, efficiency, and robustness under complex constraints have motivated research into alternative strategies. Building upon these efforts, the Artificial Potential Field (APF) method has been widely applied in UAV path planning due to its heuristic nature, computational efficiency, and ease of implementation. This method steers UAVs toward the target while avoiding obstacles by simulating attractive forces from the goal and repulsive forces from obstacles. However, the classical APF suffers from notable drawbacks: it is prone to local minima in cluttered environments, and may lead to unreachable targets or oscillatory paths in densely obstructed areas [19].

To address these issues, various enhancements have been proposed, including random perturbations, refined repulsive force functions, and virtual target settings. For example, Liu et al. [20] developed a local minimum detection mechanism that dynamically monitors path variations to accelerate the identification of local minima and optimized potential field integration to improve escape capabilities. Hao et al. [21] enhanced path accessibility by introducing virtual sub-goals. Liu et al. [22] proposed a Guided-APF approach based on environmental perception and RRT, improving object awareness in complex environments. Li et al. [23] incorporated chaos theory into the potential function to increase path diversity. Du et al. [24] introduced the Dynamic Airspace Potential Field method to enable real-time obstacle avoidance in dynamic environments. Tao [25] applied lateral forces and variable repulsion parameters to reduce the risk of UAVs being trapped in local minima. Qi et al. [26] improved path smoothness and obstacle avoidance by dynamically adjusting attractive and repulsive functions. Yan et al. [27] considered relative position, velocity, and acceleration to enhance navigation around complex-shaped obstacles. Souza et al. [28] introduced a vortex field mechanism allowing robots to autonomously choose optimal rotation directions for obstacle avoidance.

In terms of hybrid approaches, For example, the APF method can be integrated with the PSO method. The integration strategies of APF and PSO can be broadly classified into three categories. The first treats APF as a “guidance map” with PSO performing optimization within the potential field. This approach is intuitive and converges rapidly, but often relies on discretization and lacks adaptability to dynamic scenarios and continuous three-dimensional spaces [29]. The second injects APF’s attractive and repulsive trends directly into PSO’s velocity update, enabling particles to achieve global exploration while reducing collisions. This design requires little memory and can generate multiple candidate paths, but most studies remain limited to small-scale experiments without standardized large-scale comparisons [30]. The third employs a hierarchical framework in which APF secures local safety while PSO plans global waypoints or conducts replanning. Although this shows potential for online applications, it introduces higher system complexity and suffers from insufficient validation [31].

Motivated by these observations, this study adopts a distinctive fusion strategy: APF provides directional guidance, while at each planning step a localized multi-generation PSO search is performed around candidate points suggested by the potential field. The solutions from PSO are then weighted and combined with the APF solution, enabling paths to bypass obstacles from above in 3D environments with improved clearance and better engineering feasibility. Compared with the approaches of Chen et al. (Grey Wolf Optimizer-APF, GWO-APF) [32], Wan et al. (Whale Optimization Algorithm-APF, WOA-APF) [33], and Han et al. (IGWO-APF) [34], which emphasize global swarm intelligence with APF serving as a cost constraint, the proposed method achieves lower memory consumption by using smaller populations and shallower iterations, making it more suitable for onboard and real-time applications. It should be noted, however, that in maze-like environments requiring long detours and emphasizing the global shortest paths, global-search-intensive methods such as GWO/WOA/IGWO may hold an advantage. In contrast, in scenarios with dense cylindrical obstacles, constrained computational resources, and a need for online execution, the proposed method delivers more stable feasibility and superior turning characteristics at a lower computational cost, complementing GWO-APF and WOA-APF approaches. The Table 1 presents a comparative analysis of these hybrid algorithms:

3 Methodology Design

3.1 Traditional Artificial Potential Field Method

The Artificial Potential Field method was first proposed by Khatib [37]. It is a classical path planning technique grounded in the concept of force fields and has been extensively applied in mobile robotics, unmanned aerial vehicles, and related domains. This method constructs a virtual potential field within the motion space of an object, enabling the object to autonomously plan a path from the starting point to the target point under the combined effect of attractive and repulsive forces, while avoiding obstacles.

Table 1 Overview of APF—metaheuristic hybrids

Method	Core mechanism	Optimization focus	Typical application	References
APF + PSO	PSO performs global search; APF provides gradient guidance	On reported benchmarks, tends to shorten path length and improve stability	Mobile robots (static or simple dynamic obstacles)	[30, 35]
APF + GWO	APF potential coefficients are injected into the GWO hierarchical updates	Length-safety balance; mitigates local minima	Single-UAV or mobile-robot planning in complex obstacle environments	[32, 34]
APF + WOA	WOA encircling or spiral search with embedded improved potential-field factors or cascaded APF	Feasibility and path smoothness under dynamic obstacles	Industrial or outdoor dynamic scenarios	[33, 36]
Proposed method	APF-led stepwise progression + local multi-generation PSO neighborhood correction	Reduces path length while maintaining safety margins; lower online computational budget	Autonomous navigation in unknown environments	[This work]

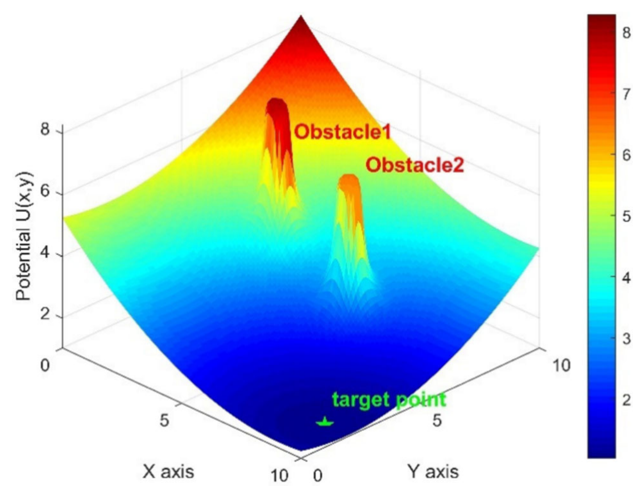
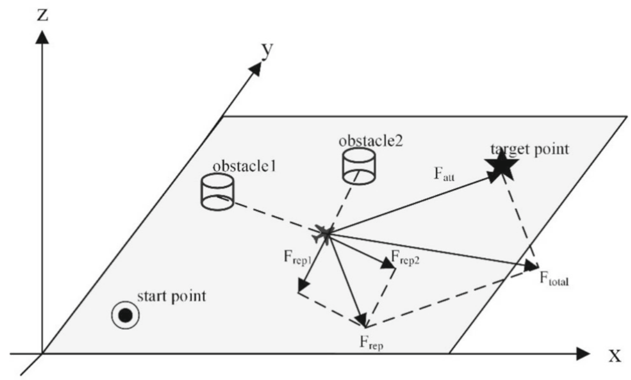


Fig. 1 Schematic diagram of the potential field

3.1.1 Principle of Potential Field Construction

In the Artificial Potential Field method, the entire environment is modeled as an artificially constructed potential field system. The target point is regarded as an attractive source, while obstacles are considered repulsive sources. A schematic diagram of the potential field is shown in Fig. 1:

Fig. 2 Schematic diagram of the potential field



The UAV is subject to an attractive force from the target point and a repulsive force from obstacles within the potential field. Its flight path is determined by the direction of the resultant force, as illustrated in Fig. 2:

The fundamental idea behind constructing the potential field can be represented by the following potential energy function [38]:

$$U\ (q) = U_{\text{att}}\ (q) + U_{\text{rep}}\ (q) \tag{1}$$

where q denotes the current position of the UAV; U_{att} represents the attractive potential exerted by the target point, and U_{rep} represents the repulsive potential exerted by the obstacles.

3.1.2 Mathematical Model

Attractive potential function The attractive potential is used to drive the UAV toward the target point and is constructed using a quadratic function [39, 40]:

$$U_{\text{att}}(q) = \frac{1}{2}k_{\text{att}} \cdot \left\| q - q_{\text{goal}} \right\|^2 \tag{2}$$

The corresponding attractive force is:

$$F_{\text{att}}\ (q) = -\nabla U_{\text{att}}\ (q) = -k_{\text{att}}\left(q - q_{\text{goal}}\right) \tag{3}$$

where k_{att} is the attractive coefficient, and q_{goal} is the position vector of the target point.

Repulsive potential function The repulsive potential is used to prevent the UAV from colliding with obstacles and is constructed in the following form:

$$U_{\text{rep}}(q) = \begin{cases} \frac{1}{2}k_{\text{rep}}\left(\frac{1}{\rho(q)} - \frac{1}{\rho_0}\right)^2, & \rho(q) \le \rho_0 \\ 0, & \rho(q) > \rho_0 \end{cases} \tag{4}$$

In the equation, $\rho(q)$ denotes the distance from the UAV to the nearest obstacle; ρ_0 is the threshold distance for the repulsive effect, beyond this distance, the repulsive force becomes inactive; and k_{rep} is the repulsive coefficient.

The corresponding repulsive force is:

$$F_{\text{rep}}(q) = \begin{cases} k_{\text{rep}} \left(\frac{1}{\rho(q)} - \frac{1}{\rho_0} \right) \frac{1}{\rho^2(q)} \cdot \nabla \rho(q), & \rho(q) \le \rho_0 \\ 0, & \rho(q) > \rho_0 \end{cases} \tag{5}$$

Resultant force calculation The total force acting on the UAV within the potential field is the sum of the attractive and repulsive forces:

$$F_{\text{total}} = F_{\text{att}} + \sum_{i=1}^{n} F_{\text{rep}}(i) \tag{6}$$

where F_{total} represents the total resultant force acting on the UAV, which determines its motion direction in the environment and guides it to find a safe path.

3.2 Problems of the Traditional Artificial Potential Field Method

The traditional Artificial Potential Field method mainly suffers from two issues: one is the local minimum problem, and the other is the unreachable target problem.

3.2.1 Local Minimum Problem

The local minimum problem in the traditional Artificial Potential Field method mainly stems from the fact that both the attractive and repulsive potential functions are constructed as continuous and smooth functions. As a result, the combined potential field may contain multiple local minima. When the UAV reaches a position where the magnitudes of the attractive and repulsive forces are equal, the resultant force becomes zero. If this equilibrium point is not the target location, the UAV becomes trapped in a local minimum [41].

If the UAV passes through such a local minimum during flight, it may hover near the point or come to a complete stop, leading to failure in path planning. The local minimum problem is defined as follows:

Let q denote the current position of the UAV. The total potential energy function $U(q)$ is defined as in Eq. (1).

If there exists a point $q* \neq q_{\text{goal}}$ such that: $\nabla U(q*) = 0, \forall \varepsilon > 0, U(q) < U(q + \varepsilon)$, then $q*$ is defined as a local minimum point. If the UAV passes through a local minimum during its flight, it will repeatedly hover near that point or remain stationary, ultimately causing path planning to fail.

Typical examples of local minimum problems are illustrated in Figs. 3 and 4. In Fig. 3, the UAV, the target point, and the obstacle lie on the same straight line. Therefore, there must exist a point along this line where the attractive and repulsive forces are equal in magnitude but opposite in direction, causing the UAV to become trapped in a local minimum. In Fig. 4, a narrow passage is formed between obstacle 1 and obstacle 2. The combined repulsive force generated by both obstacles counterbalances the attractive force from the target point, resulting in the formation of a local minimum point within the passage.

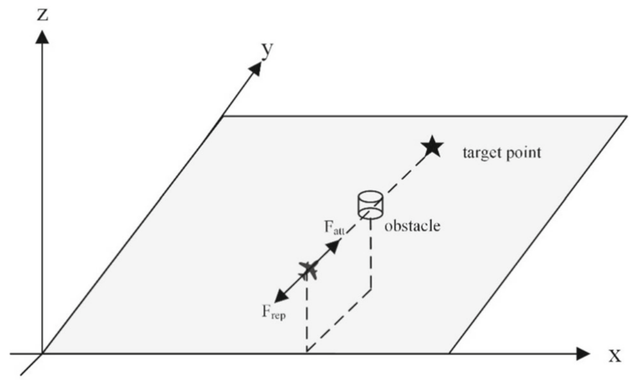


Fig. 3 Force analysis diagram of the local minimum problem

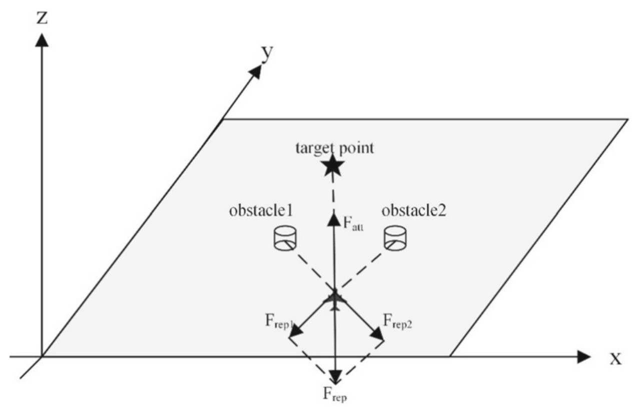


Fig. 4 Force analysis diagram of the local minimum problem

3.2.2 Unreachable Target Problem

In the traditional Artificial Potential Field method, if there are obstacles located near the target point, the UAV's distance to the obstacles decreases as it approaches the target. According to Eqs. (3) and (5), the attractive force between the UAV and the target point is proportional to their distance. As the distance decreases, the attractive force weakens. Conversely, within the influence region of an obstacle, the repulsive force is inversely proportional to the distance between the UAV and the obstacle, thus, as the distance decreases, the repulsive force increases.

When the UAV gets very close to both the target point and nearby obstacles, the repulsive force may exceed the attractive force, preventing the UAV from reaching the target. This phenomenon is referred to as the unreachable target problem, as illustrated in Fig. 5. The description of the unreachable target problem is as follows: When the target point $q_{\text{goal}} \in D_{\text{rep}}$, $D_{\text{rep}} = \{\{q \in R^n \mid \rho(q) \leq \rho_0\}$ lies within the effective repulsive field region, and there exists an arbitrary path $s(t)$ such that:

$$\lim_{t \to \infty} \left\| s(t) - q_{\text{goal}} \right\| \neq 0 \tag{7}$$

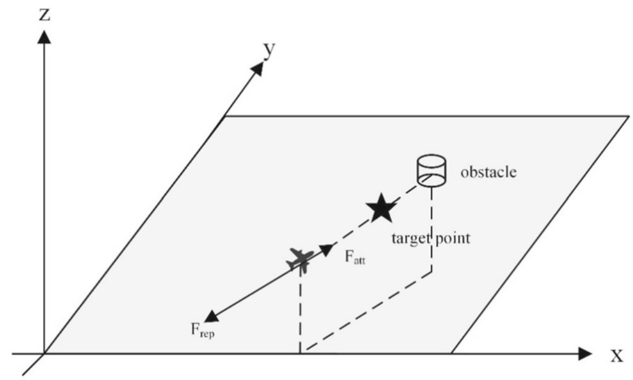
It indicates that the UAV cannot converge to the target point, i.e., the target is unreachable.

3.3 Improved Potential Field Method Based on Particle Swarm Optimization

3.3.1 Gaussian Repulsive Function

In the traditional Artificial Potential Field method, the repulsive force component lacks smoothness at the boundary of the obstacle's influence range. When the UAV enters this range, a large repulsive force is applied abruptly, whereas outside the range, the repulsive force drops to zero. This discontinuity at the boundary causes the UAV to oscillate or jitter near obstacles.

Fig. 5 Force analysis diagram of the unreachable target problem



To address this issue, we introduce a Gaussian repulsive function, which enhances the local responsiveness of the potential field guidance, reduces interference from repulsive forces near the target, and improves the adaptability of the planned path to complex terrain [42]. The Gaussian function, originally used in signal processing, is commonly applied to suppress noise and limit interference range.

The Gaussian repulsive function is defined as follows:

$$F_{\text{rep}}(q) = K_{\text{rep}} \cdot \exp\left(-\frac{(\rho(q)-\zeta)^2}{2\sigma^2}\right) \cdot \left(\left(\frac{1}{\rho(q)} - \frac{1}{\zeta}\right)\frac{1}{\rho^2(q)} \cdot \vec{d}\right) \tag{8}$$

where ζ denotes the effective influence range, σ is the Gaussian width parameter, and $\vec{d}$ is the unit vector pointing from the obstacle to the UAV.

The closer the UAV is to the obstacle, the greater the repulsive force;conversely, the farther it is, the weaker the force. By using the Gaussian repulsive function, the planned path can maintain smooth transitions even near the edges of obstacles, and the influence of distant obstacles on the path is significantly reduced.

Figure 6 shows a comparison between the Gaussian repulsive force and the traditional repulsive force:

Figure 6 illustrates the variation in normalized repulsive force intensity between the classical repulsive force model and the improved model incorporating a Gaussian modulation factor, under different distances from an obstacle. The horizontal axis represents the distance to the obstacle center, while the vertical axis indicates the normalized repulsive force. As shown in the figure, the Gaussian repulsive force decreases more gradually and smoothly compared to the traditional repulsive force.

3.3.2 Fusion Particle Swarm Optimization and Artificial Potential Field Method

Particle swarm optimization Particle Swarm Optimization is a population-based metaheuristic algorithm originally introduced by Kennedy and Eberhart [43]. Inspired by the social behavior of biological populations such as bird flocks and fish schools, PSO simulates the collaborative search process of individuals (particles) in a high-dimensional solution space. Each particle i in a D-dimensional space maintains a position vector $\boldsymbol{x}_i = [x_{i1}, x_{i2}, \ldots, x_{iD}]$, a velocity vector $\boldsymbol{v}_i = [v_{i1}, v_{i2}, \ldots, v_{iD}]$, and its personal best position $\boldsymbol{p}_i$. The swarm also tracks the global best position $\boldsymbol{g}$ found so far.

At each iteration, particle i updates its velocity and position as follows:

$$\boldsymbol{v}_i(t+1) = w \cdot \boldsymbol{v}_i(t) + c_1 \cdot r_1 \cdot (\boldsymbol{p}_i - \boldsymbol{x}_i(t)) + c_2 \cdot r_2 \cdot (\boldsymbol{g} - \boldsymbol{x}_i(t)) \tag{9}$$

$$\boldsymbol{x}_i(t+1) = \boldsymbol{x}_i(t) + \boldsymbol{v}_i(t+1) \tag{10}$$

Where w denotes the inertia weight that balances exploration and exploitation, c_1 and c_2 are acceleration coefficients representing cognitive and social components, respectively, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are random variables sampled from a uniform distribution.

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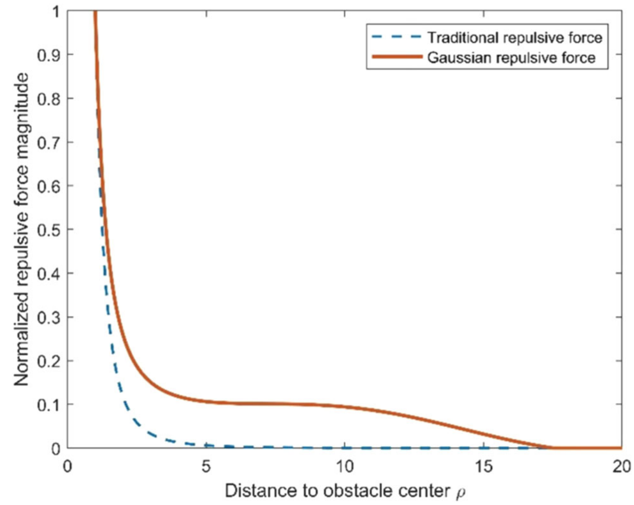


Fig. 6 Comparison between the Gaussian repulsive force and traditional repulsive force

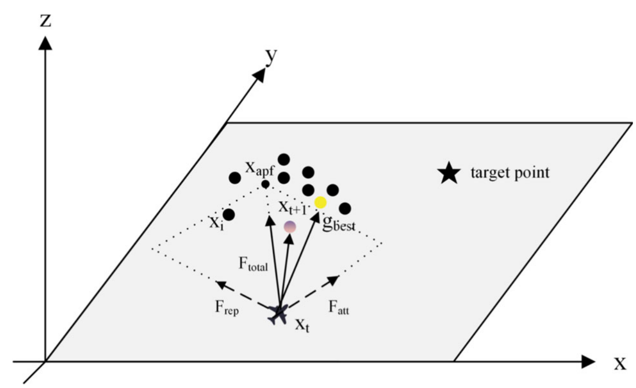


Fig. 7 Schematic diagram of the PSO-APF hybrid method

Due to its simplicity, ease of implementation, and strong global optimization capability, PSO has been widely applied to various complex optimization problems, such as function optimization, path planning, neural network training, and control systems [44, 45].

Fusion mechanism design This paper proposes a hybrid path planning strategy that integrates the APF method with the PSO algorithm. In this approach, APF provides directional guidance by generating an initial path through a force field model, while PSO refines and optimizes the path points to enhance obstacle avoidance and ensure path feasibility.

The integration process is illustrated in Fig. 7 and can be divided into the following three stages:

(1) Calculation of the guidance direction using the Artificial Potential Field:

First, based on the current position of the UAV, the total resultant force is calculated using the potential field method by combining the attractive and repulsive forces. This resultant force serves as the reference direction for the UAV's next movement. The attractive force vector is derived using a linear attraction model:

$$F_{\text{att}} = K_{\text{att}} \cdot \left(x_{\text{goal}} - x_t\right) \tag{11}$$

where x_t represents the current position of the UAV; and x_{goal} denotes the target position.

The repulsive force is defined via a Gaussian potential function, as shown in Eq. (8). Accordingly, the resultant force is computed as the combination of attractive and repulsive terms:

$$F_{\text{total}} = F_{att} + F_{\text{rep}} \tag{12}$$

At each path point, the resultant force vector is first constructed by summing the attractive and repulsive forces. Based on this vector, the potential field guidance point is calculated as the position influenced by the resultant force:

$$x_{\text{apf}} = x_t + F_{\text{total}} \tag{13}$$

(2) Local refinement using Particle Swarm Optimization:

Around the guidance point x_{apf} generated by the resultant force, a particle swarm is constructed to perform local search and optimization. First, particle initialization is conducted: taking x_{apf} as the center, N particles are initialized, and random perturbations are introduced. The position of each particle x_i is determined by:

$$x_i = x_{\text{apf}} + \delta_i, \quad \delta_i \sim \mathcal{N}(0, \sigma^2 I) \tag{14}$$

where δ_i is a three-dimensional perturbation vector drawn from a zero-mean Gaussian distribution with a covariance matrix $\sigma^2 I$, indicating that each dimension is independently and identically distributed.

Each particle is evaluated using the following fitness function:

$$f(x_i) = \left\| x_i - x_{\text{goal}} \right\| + \sum_{k=1}^{N} P_k(x_i) \tag{15}$$

where, $P_k(x_i)$ denotes the penalty term related to the proximity of particle x_i to the k-th obstacle.

Each particle maintains its personal best position p_i^{best} throughout the iterations. The global best position g_{best} is selected as the one with the lowest fitness value among all p_i^{best}:

$$p_i^{\text{best}} = \arg \min_{t \in [1,T]} f(x_i^t) \tag{16}$$

$$g_{\text{best}} = \arg \min_{i} f(p_i^{\text{best}}) \tag{17}$$

(3) Final linear fusion for each path point:

$$x_{t+1} = \alpha \cdot x_{\text{apf}} + (1 - \alpha) \cdot g_{\text{best}} \tag{18}$$

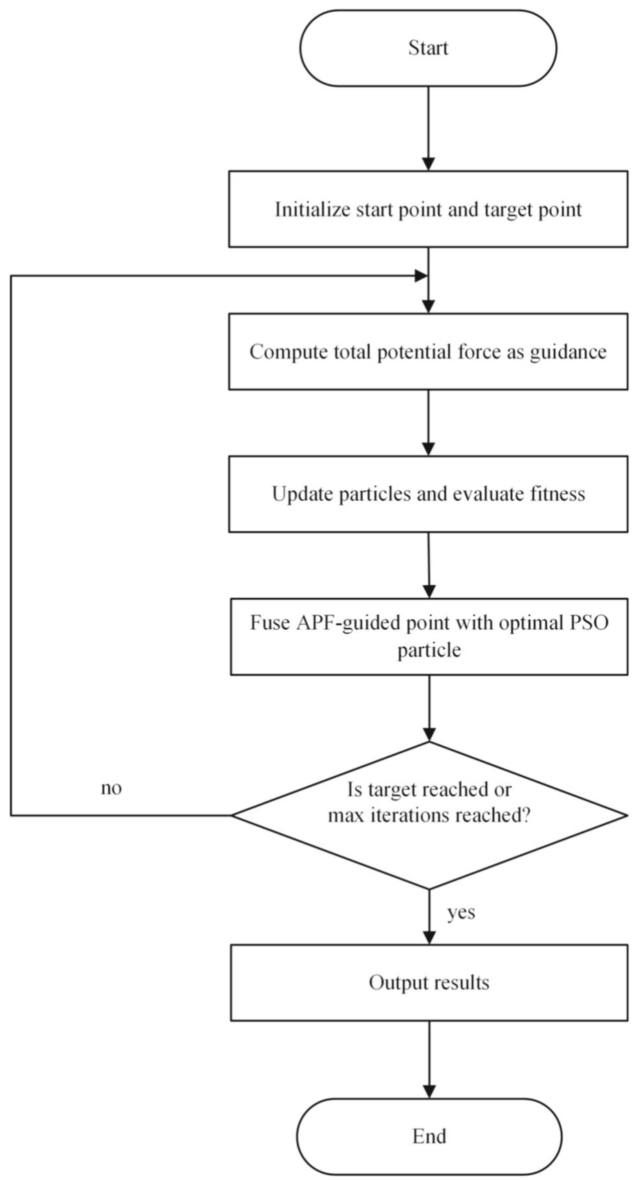
where, x_{t+1} represents the next position of the UAV; α is the fusion coefficient used to balance the global guidance of the artificial potential field method and the search capability of the particle swarm optimization algorithm.

By introducing the particle swarm optimization mechanism, local path refinement can be performed at each step following the guidance of the artificial potential field, thereby enhancing the global exploration and local adaptability of the path planning algorithm. This hybrid method combines the strengths of the APF method, such as high computational efficiency and clear directional guidance, with the ability of PSO to escape local minima in complex search spaces, significantly improving the algorithm's adaptability to complex environmental structures.

Compared to the traditional APF, which is prone to getting trapped in local minima or encountering unreachable target issues in densely obstructed environments, the integration of PSO enables the generation of multiple search particles in the local region. These particles undergo fitness evaluation, effectively avoiding potential field traps and exploring more optimal paths. This significantly enhances the robustness and flexibility of the path planning process.

In multi-obstacle environments, the hybrid method demonstrates greater fault tolerance and global convergence capability, offering a more robust solution for the UAV path planning in three-dimensional space (Fig. 8). The pseudocode is defined as follows:

Fig. 8 Flowchart of the FPAPF algorithm



Algorithm 1 The FPAPF pseudocode

Input: Start point $\mathbf{s}$, target point $\mathbf{g}$, max iterations $T_{\max}$, threshold distance ϵ
Initialize: $\mathbf{p}_0 \leftarrow \mathbf{s}$, iteration counter $t \leftarrow 0$
while $\|\mathbf{p}_t - \mathbf{g}\| > \epsilon$ **and** $t < T_{\max}$ **do**
  Compute force: $\mathbf{F}_{\text{total}} \leftarrow \mathbf{F}_{\text{att}} + \mathbf{F}_{\text{rep}}$
  Guidance point: $\mathbf{p}_{apf} \leftarrow \mathbf{p}_t + \mathbf{F}_{\text{total}}$
  Initialize N particles around $\mathbf{p}_{apf}$
  Set initial personal bests $p_i^{best} \leftarrow x_i$, and global best g_{best}
  for $k = 1$ to K **do**
    for each particle i **do**
      Update velocity and position of x_i
      Evaluate fitness $f(x_i)$
      if $f(x_i) < f(p_i^{best})$ **then**
        Update $p_i^{best} \leftarrow x_i$
      end if
      if $f(x_i) < f(g_{best})$ **then**
        Update $g_{best} \leftarrow x_i$
      end if
    end for
  end for
  Compute new position: $\mathbf{p}_{t+1} \leftarrow \alpha \cdot \mathbf{p}_{apf} + (1-\alpha) \cdot \mathbf{g}_{best}$
  $t \leftarrow t + 1$
end while
Return Results

3.4 Path Evaluation Metric Design

3.4.1 Evaluation Metric Formulation

To comprehensively assess the overall performance of different path planning algorithms in a three-dimensional environment, this study designs a weighted composite scoring metric based on three key factors: path length (Length), average safety distance (Avg. safety), and minimum safety distance (Min. safety). This metric provides a unified framework for evaluating the performance of each algorithm in terms of both safety and efficiency.

First, the path length L reflects the overall efficiency of the path planning algorithm, smaller values indicate shorter paths and more optimal flight distances. Second, the average safety distance D_{avg} measures the average spatial clearance between the path and surrounding obstacles, a larger value implies that the path is generally safer. Additionally, the minimum safety distance D_{min} captures the most critical or hazardous point along the path, a higher value indicates a greater likelihood of successful obstacle avoidance by the algorithm.

To enable dimensionless comparison, the three evaluation metrics are normalized individually. Among them, path length is normalized in an inverse manner, such that shorter paths receive higher scores. The normalization formulas are defined as follows:

$$\text{score} = \omega_1 \cdot \left(1 - \frac{L}{L_{\max}}\right) + \omega_2 \cdot \left(\frac{D_{\text{avg}}}{D_{\text{avg,max}}}\right) + \omega_3 \cdot \left(\frac{D_{\min}}{D_{\text{min,max}}}\right) \tag{19}$$

where $L_{\max}$, $D_{\text{avg,max}}$ and $D_{\text{min,avg}}$ represent the maximum values of the corresponding metrics among all algorithms. $\omega_1, \omega_2, \omega_3$ are the weighting coefficients, subject to the constraint $\omega_1 + \omega_2 + \omega_3 = 1$.

3.4.2 Derivation of Weights via the Analytic Hierarchy Process

Analytic Hierarchy Process (AHP) [46] vis a multi-criteria decision-making and weighting methodology: it decomposes a problem into a goal-criteria-alternatives hierarchy, uses pairwise comparisons to quantify relative importance into coefficients, and employs a consistency test to ensure non-contradictory judgments. Liu et al. [47] in a travel time–cost setting used Saaty's 1–9 scale to construct the pairwise comparison judgment matrix, then derived the weights following standard AHP procedures, and explicitly performed a consistency check. Tang et al. [48] determined risk-factor weights for a minimum-risk cost objective via AHP, detailing the hierarchical modeling, the nine-level scale, and the geometric-mean method for weight extraction, and computed the consistency ratio (CR) before incorporating the weights into the cost function. Liu et al. [49] likewise estimated factor weights for a multi-UAV cost-benefit function using AHP and provided the AHP scale table as the basis for judgments.

In sum, AHP is readily applicable to UAV path-planning problems, establishing the paradigm pairwise comparisons through weight derivation and consistency validation. Accordingly, AHP is applied to the three criteria (path length, average safety distance, minimum safety distance); pairwise comparisons are performed and, following a consistency check, the resulting weights are used for the composite score.

The steps of the AHP are as follows:

(1) Pairwise comparisons. Using Saaty's 1–9 scale to assess the importance of criterion i relative to criterion j, construct the judgment matrix:

$$A = \left[a_{ij}\right], \quad a_{ij} = \frac{1}{a_{ji}}, \ a_{ii} = 1$$

The AHP rating scale is given below:

(2) Criterion weights were derived via the principal right-eigenvector method. Specifically, solve:

$$Av = \lambda_{\max} v \tag{20}$$

and then normalize:

$$\omega_i = \frac{v_i}{\sum_{k=1}^{n} v_k}, \quad i = 1, \ldots, n. \tag{21}$$

(3) A consistency check is performed to ensure that the pairwise comparisons are logically coherent. The consistency index (CI) and consistency ratio (CR) are given by:

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n - 1}, \qquad \mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}(n)} \tag{22}$$

where $\mathrm{RI}(n)$ is the random index. Common values are:

$$RI(1, \ldots, 10) = [0, 0, 0.58, 0.90, 1.12, 1.24, 1.32, 1.41, 1.45, 1.49].$$

A CR< 0.10 is conventionally regarded as indicative of acceptable consistency. Accordingly, AHP is employed to derive criterion weights for the composite evaluation of path-planning schemes. The criteria are the path length L, the average safety distance D_{avg}, and the minimum safety distance $D_{\min}$. Under a task specification that prioritizes efficiency while accounting for safety, pairwise comparisons among the three criteria are conducted using the Saaty 1–9 scale (Table 2), with the following judgment rationales:

L over D_{avg}:4 (between moderate and strong). In tasks with stringent real-time or energy constraints, shortening the path length exerts a more direct and sustained impact on mission completion time and energy consumption.

L over $D_{\min}$:7 (very strong). The minimum clearance primarily serves as a pass fail feasibility safeguard; once it exceeds the engineering threshold, the marginal benefit diminishes rapidly.

D_{avg} over $D_{\min}$: 3 (moderate). After the baseline feasibility is satisfied, increasing the overall clearance reduces risk along the entire trajectory.

Table 2 Saaty's 1–9 scale for pairwise comparisons

Intensity of importance	Definition
1	Equal importance
3	Slight importance
5	Moderate importance
7	Strong importance
9	Extreme importance
2, 4, 6, 8	Intermediate importance between adjacent odd levels

Accordingly, the judgment matrix is:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 1/4 & 1 & 3 \\ 1/7 & 1/3 & 1 \end{bmatrix}.$$

Solving for the principal eigenpair yields $\lambda_{\max} = 3.032$ with eigenvector:

$$v_1 = [0.9518,\ 0.2847,\ 0.1136]^{\mathrm{T}},$$

which, after normalization, gives the weight vector:

$$\omega = (0.705,\ 0.211,\ 0.084).$$

The consistency index is:

$$\mathrm{CI} = \frac{\lambda_{\max} - 3}{2} = 0.016,$$

and with $\mathrm{RI}(3) = 0.58$, the consistency ratio is:

$$\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}} = 0.028 < 0.10,$$

indicating acceptable consistency.

Following the AHP analysis, the criterion weights $(0.705,\ 0.211,\ 0.084)$ are rounded to $(0.7,\ 0.2,\ 0.1)$ for use in Eq. (19). This rounding is adopted for ease of implementation.

To balance efficiency and safety, Eq. (19) combines the inverse-normalized path length with the dimensionless indicators of average and minimum safety distances in a linear fashion, resulting in a unified score ranging from [0,1]. This facilitates direct comparison and ranking across different algorithms and scenarios. The use of both average and minimum safety distances reflects the overall safety margin while also capturing the risk at the most vulnerable locations, thereby avoiding bias that may arise from using a single safety indicator. The weights are derived through the standard procedure of the Analytic Hierarchy Process and validated via consistency checks, ensuring transparency and reproducibility in the evaluation. This scoring method is simple, low-cost, and suitable for large-scale experiments and statistical validation, and can be integrated with engineering hard constraints.

3.5 Path Executability Verification Method

Although path planning algorithms are capable of generating obstacle-avoiding trajectories in three-dimensional space, the actual executability of such paths largely depends on the flight control system's ability to effectively track the planned trajectory. In complex environments, the execution performance is influenced not only by the geometric smoothness and rationality of the path, but also by a combination of factors, including controller performance, external disturbances, and perception errors [50]. To comprehensively assess the feasibility and stability of the planned path under realistic conditions, we propose a controller simulation-based verification framework that emulates the execution process of unmanned aerial vehicles in real-world scenarios. This framework supports the switching and comparison of multiple representative control strategies, and under a unified trajectory input and disturbance modeling scheme, reveals the potential risks that may arise during execution from the perspective of control response.

Through this method, path executability is no longer limited to offline geometric analysis under ideal conditions, but is instead systematically verified within a dynamic closed-loop system that incorporates environmental disturbances and control feedback. The proposed verification approach serves as a bridging mechanism between path planning algorithms and flight control systems, with a focus on the dynamic realizability of planned trajectories in a closed-loop control context. Compared to conventional offline evaluations that rely solely on geometric features such as curvature and jerk, this approach provides a more comprehensive and realistic assessment by modeling system dynamics, controller behavior, and environmental perturbations during actual execution.

3.5.1 Verification Objectives and Design Principles

To evaluate the trackability and stability of different controllers in executing planned trajectories, especially under practical challenges such as non-smooth paths and modeling uncertainties, a dedicated validation framework is developed. To enhance the applicability and referential value of the proposed method, the system is designed based on the following principles: (1) providing seamless controller switching and supporting multiple control strategies; (2) incorporating observation errors and system variations to better simulate real-world conditions; and (3) constructing a multi-dimensional performance evaluation system that comprehensively assesses tracking accuracy, smoothness, and robustness. The trajectory executability validation process is divided into five key stages: trajectory input, controller integration, flight system simulation, environmental disturbance and error modeling, and performance evaluation with executability analysis. The overall validation workflow is illustrated in Fig. 9.

3.5.2 System Architecture and Control Model

The validation method is implemented on a modular simulation platform that supports the execution of multiple control strategies within a unified framework, thereby enabling systematic comparative performance analysis. The system takes a sequence of 3D waypoints as input, simulates the state evolution of the UAV during trajectory tracking tasks, and outputs results such as the tracked trajectory and control logs.

The controlled object is modeled as a point mass in three-dimensional space, whose motion is described by position and velocity states in discrete time. The control input is defined as acceleration. The system states are updated at fixed time intervals according to the following motion equations:

$$\begin{aligned} v_{k+1} &= v_k + u_k \cdot \Delta t \\ x_{k+1} &= x_k + v_{k+1} \cdot \Delta t \end{aligned} \tag{23}$$

where x_k and v_k represent the position and velocity at the current time step, u_k denotes the acceleration control input, and Δt is the simulation time step.

To evaluate the executability of planned paths at the control level, a simulation platform with controller switching capability is developed. The system takes a sequence of trajectory waypoints as input and outputs three-dimensional acceleration commands as control inputs to drive the UAV along the path. Considering the characteristics of different control strategies, three representative controllers are selected for comparison: a Linear Quadratic Regulator (LQR) based on linear state feedback [51], a conventional Proportional-Integral-Derivative (PID) controller [52], and an Enhanced LQR (ELQR) controller proposed in this work.

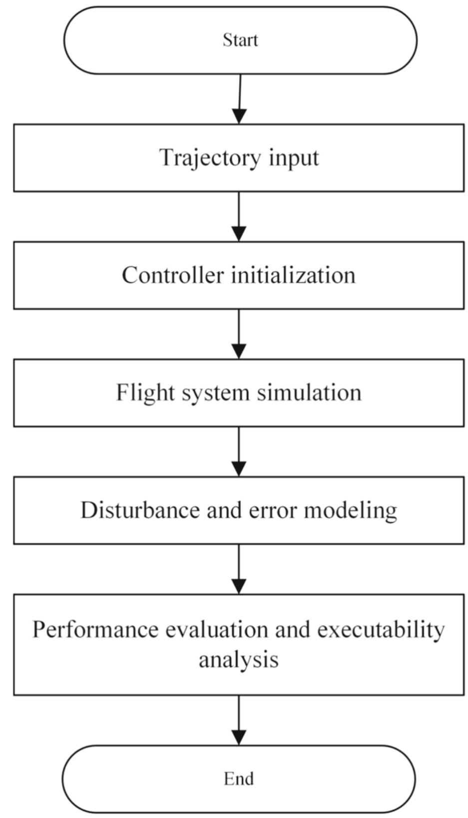


Fig. 9 Flowchart of the trajectory tracking validation process

(1) LQR Controller

A linear second-order system model is used to establish the state feedback control law. For each spatial dimension, the state is defined as $X_k = \left[x_k,\ v_k\right]^{\top}$. The standard state-space feedback is constructed as follows:

$$u_k = -K \cdot \left(X_k - X_k^{\mathrm{ref}}\right) \tag{24}$$

where $X_k{}^{\mathrm{ref}} = \left[x_k{}^{\mathrm{ref}}, v_k{}^{\mathrm{ref}}\right]^{\mathrm{T}}$ represents the reference state vector at step k, including the reference position x_k^{ref} and reference velocity v_k^{ref}. The gain matrix K is obtained offline by solving a discrete-time LQR cost minimization problem, which balances trajectory tracking error and control energy consumption. This method provides good stability and theoretical guarantees, making it suitable for tracking trajectories of moderate complexity.

(2) PID Controller

The PID controller is a classical feedback control method that stabilizes the system response by separately reacting to the present value, accumulated history, and the rate of change of the tracking error through its proportional, integral, and derivative terms. Its discrete form is given by:

$$\begin{aligned} u_k &= K_p \cdot e_k + K_i \sum_{j=0}^{k} e_j \cdot \Delta t + K_d \cdot \frac{e_k - e_{k-1}}{\Delta t} \\ e_k &= x_k^{\mathrm{ref}} - x_k \end{aligned} \tag{25}$$

where e_k denotes the current position tracking error; K_p, K_i, and K_d represent the proportional, integral, and derivative gains, respectively.

(3) ELQR Controller

To improve the tracking performance of the system in non-smooth trajectory segments, an enhanced version of the standard LQR controller is proposed. In addition to the state feedback, this controller incorporates a velocity feedforward compensation mechanism. By performing a look-ahead along the planned path, the expected velocity of future target points is predicted to dynamically adjust the control input. The corresponding control strategy can be expressed as:

$$\begin{aligned} u_k &= -K \cdot \left(X_k - X_k^{\text{ref}}\right) + \alpha \cdot \left(\hat{v}_k - v_k\right) \\ \alpha &= \alpha_0 + \frac{\alpha_1}{1 + \left\| X_k - X_k^{\text{ref}} \right\|} \\ \hat{v}_k &= \frac{x_{k+L}^{\text{ref}} - x_k^{\text{ref}}}{L \cdot \Delta t} \end{aligned} \tag{26}$$

where α denotes an adaptive gain associated with the tracking error, which adjusts the control intensity based on the deviation magnitude. The parameters α_0 and α_1 are tunable coefficients. $\dot{v}_k^{\text{ref}}$ represents the predicted reference velocity change obtained by looking ahead L steps along the planned path from the current position. This mechanism enhances the responsiveness and anticipatory capability of the controller under complex trajectories, sharp turns, or sudden maneuvers, thereby significantly improving tracking accuracy and robustness.

3.5.3 Environment Modeling and Error Simulation

To better approximate real flight conditions, the system incorporates representative disturbance factors into the trajectory tracking validation process to capture the uncertainties encountered by UAVs during actual operation. These factors mainly include observation errors, system parameter variations, and external disturbances. Observation errors are used to simulate measurement inaccuracies during sensor readings, and are modeled by adding random noise to the position measurements at each control cycle. System parameter variations are implemented by adjusting the UAV's mass within a defined range, capturing the effects of payload fluctuations, battery level changes, and their influence on system dynamics. External disturbances represent uncontrollable environmental factors, such as wind or turbulent airflow, that affect control performance. These are simulated by superimposing small random perturbations on the control inputs.

Through the modeling of these disturbances, the system enables the evaluation of control strategies in near-realistic conditions, allowing for a unified operating environment to assess the accuracy, stability, and adaptability of each controller.

3.5.4 Control Performance Evaluation Metrics

To comprehensively evaluate the tracking performance of various control strategies under different trajectories and disturbance conditions, the system defines multiple performance metrics. These metrics assess the controllers quantitatively from several perspectives, including trajectory error, dynamic smoothness, and system stability. The evaluation is based on the deviation between the actual trajectory generated by the controller and the reference trajectory, as well as the variation in control inputs over time (Table 3). The main evaluation metrics are categorized as follows:

All the above metrics are automatically computed through a unified data logging module, and the final results are presented in a structured table format to facilitate subsequent analysis and comparison between controllers.

Table 3 Primary evaluation metrics

Metric	Description
RMSE	Average deviation over the tracking process; reflects control accuracy
Max_err	Maximum single point error during execution; indicates peak deviation
Final_err	Distance to the target at the end; evaluates steady state performance
Mean_jerk	Average rate of acceleration change; indicates motion smoothness
Max_jerk	Maximum jerk; reflects abruptness in motion
Robustness	Inverse of error variance; measures system stability
DAR	Disturbance-to-error ratio; quantifies disturbance rejection ability

4 Simulation Experiments

4.1 Algorithm Comparison

To validate the effectiveness of the proposed UAV 3D path planning method based on the fusion of Particle Swarm Optimization and Artificial Potential Field, simulation experiments were conducted in the MATLAB R2018b environment. In this study, the fusion Particle Swarm Optimization and Artificial Potential Field (FPAPF) is evaluated against five existing approaches, namely: the Traditional Artificial Potential Field method (TAPF), the Rapidly-exploring Random Tree method (RRT), the Improved Artificial Potential Field algorithm [25] (IAPF), the Modified Artificial Potential Field algorithm [28] (MAPF), and the integration of Grey Wolf algorithm and Artificial Potential Field [32] (GWOAPF).

The potential field parameters used in all experiments were fixed as follows: the repulsive coefficient was set to $K_{\text{rep}} = 0.10$, and the attractive coefficient to $K_{\text{att}} = 0.04$. The maximum number of iterations for TAPF, IAPF, MAPF, GWOAPF, and FPAPF was set to 1000. For RRT, the iteration limit was increased to 50,000 because a lower setting (e.g., 1000) frequently failed to produce feasible paths. This higher iteration count ensures the feasibility of path generation in complex environments. Within the experimental framework, the algorithms are governed by the following constraints:

1. The workspace is bounded, with both planning and sampling constrained within a fixed three-dimensional region.

$$x \in [0, 200], \;\; y \in [0, 200], \;\; z \in [0, 100]$$

2. Impenetrable obstacles: for each cylindrical obstacle with center $c_j = (x_j, y_j)$, radius r_j, and height h_j, any point $p = (x, y, z)$ must satisfy:

$$\|p_{xy} - c_j\| \geq r_j \quad \text{or} \quad z > h_j$$

otherwise, the solution is considered infeasible.

3. The repulsive effect has a limited scope, being effective only when the point is located below the cylinder top and sufficiently close to its lateral surface.

$$z < h_j, \quad \|p_{xy} - c_j\| \leq \xi_j, \quad \xi_j = 3.5 r_j$$

4. APF-PSO integration constraints: local search is performed by PSO in the vicinity of candidate positions suggested by the artificial potential field; particles that fall into or approach obstacles are either heavily penalized in the fitness function or directly discarded, ensuring that both the search process and the resulting paths remain within the feasible domain.

 https://doi.org/10.1007/s44196-025-01009-w

Table 4 The information of threat areas

Environment no.	Start	Target	Obstacles
Environment 1	(10, 10, 0)	(185, 120, 60)	8
Environment 2	(10, 10, 0)	(160, 80, 100)	10
Environment 3	(10, 10, 0)	(165, 180, 80)	12

Table 5 Coordinate information of obstacles in Environment 1

Obstacle no.	Center (x, y)	Height	Radius
Obstacle 1	(70,50)	60	4
Obstacle 2	(75,80)	60	9
Obstacle 3	(130,60)	50	7
Obstacle 4	(25,25)	60	5
Obstacle 5	(100,48)	33	6
Obstacle 6	(150,110)	35	9
Obstacle 7	(110,80)	20	5
Obstacle 8	(90,40)	20	4

Table 6 Path data of each algorithm in Environment 1

Algorithm	Success	Time (s)	Length (m)	Avg. safety (m)	Min. safety (m)
TAPF	Yes	19.69	363	14.06	2.61
RRT	Yes	18.56	293	25.93	0.10
IAPF	Yes	4.14	320	19.33	3.99
MAPF	Yes	49.57	377	7.96	0.02
GWOAPF	Yes	4.20	351	12.81	1.56
FPAPF	Yes	3.41	236	12.42	2.38

In summary, all algorithms operate under a unified constraint framework to guarantee the fairness of experimental comparisons.

4.1.1 Experimental Analysis in Static Environments

To evaluate the path-planning performance of the proposed fusion algorithm in static environments, three scenarios with different densities of cylindrical obstacles were constructed. The parameters of each environment are listed in Table 4.

Environment 1 Table 5 presents the coordinate information of each obstacle in Environment 1:

In Environment 1, six algorithms were tested. The path data for each algorithm are presented in Table 6, while the top view and 3D view of the paths generated by each algorithm are shown in Figs. 10 and 11, respectively.

In terms of path efficiency, the total length of the path generated by FPAPF is 236 m, the shortest among the six algorithms. Compared to TAPF, RRT, IAPF, MAPF, and GWOAPF, the path length is reduced by 35%, 19%, 26%, 37%, and 33%, respectively. Regarding computation time, FPAPF requires only 3.41 s, significantly faster than TAPF and MAPF, and slightly outperforming IAPF and GWOAPF.

Regarding path safety, the algorithms exhibit more pronounced differences. RRT achieves an average safety distance of 25.93 m, indicating generally large clearance from obstacles. However, its minimum safety distance is only 0.10 m, with some segments coming dangerously close to obstacles, posing potential risks. IAPF achieves the largest minimum safety distance, offering a more robust path, but at the cost of detours and lower path efficiency.

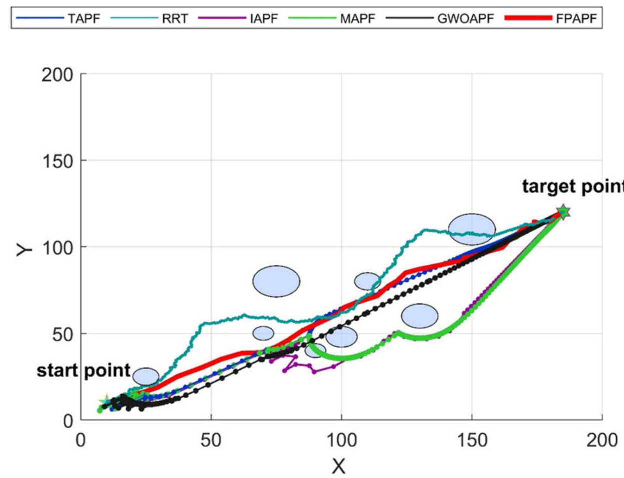


Fig. 10 Top view of paths generated by each algorithm in Environment 1

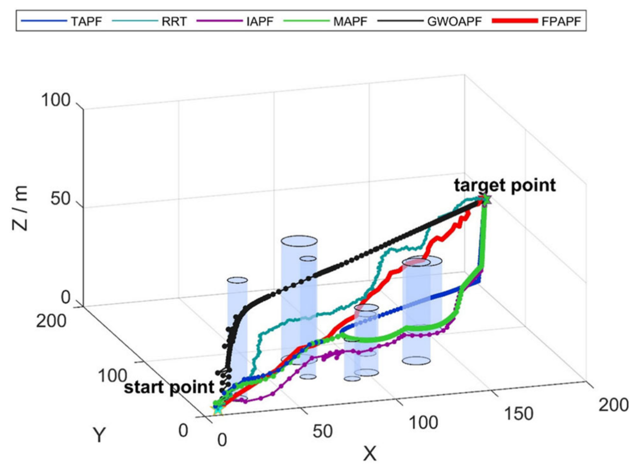


Fig. 11 3D view of paths generated by each algorithm in Environment 1

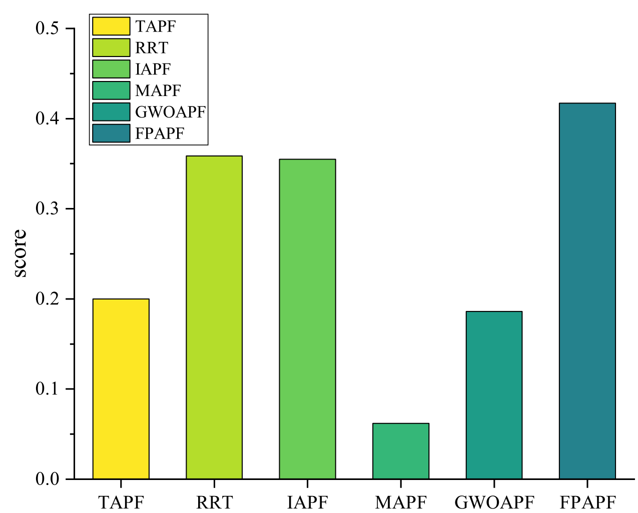


Fig. 12 Weighted scores of algorithms in Environment 1

In contrast, FPAPF maintains a compact path while achieving an average safety distance of 12.42 m and a minimum safety distance of 2.38 m, without any segments dangerously close to obstacles.

To balance the trade-off between path efficiency and safety, a weighted scoring model was employed in this study. Three core evaluation metrics were normalized and used to compute an overall performance score. As shown in Fig. 12, FPAPF achieved the highest comprehensive score, significantly outperforming the other five algorithms.

Table 7 Coordinate information of obstacles in Environment 2

Obstacle no.	Center (*x*, *y*)	Height	Radius
Obstacle 1	(70, 50)	45	6
Obstacle 2	(80, 40)	70	4
Obstacle 3	(130, 60)	70	7
Obstacle 4	(25, 25)	60	5
Obstacle 5	(150, 50)	50	4
Obstacle 6	(100, 48)	33	6
Obstacle 7	(185, 110)	35	9
Obstacle 8	(110, 80)	40	5
Obstacle 9	(90, 40)	30	4
Obstacle 10	(180, 50)	30	5

Table 8 Path data of each algorithm in Environment 2

Algorithm	Success	Time (s)	Length (m)	Avg. safety (m)	Min. safety (m)
TAPF	Yes	41.44	323	9.32	3.31
RRT	Yes	22.44	281	22.72	0.12
IAPF	Yes	3.19	289	19.03	4.05
MAPF	Yes	52.34	298	7.16	3.65
GWOAPF	Yes	3.12	225	13.50	0.36
FPAPF	Yes	3.06	210	10.35	2.03

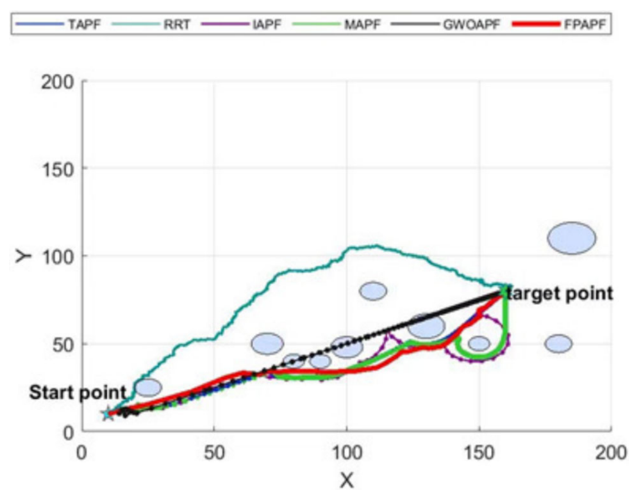


Fig. 13 Top view of paths generated by each algorithm in Environment 2

Environment 2 Table 7 presents the coordinate information of each obstacle in Environment 2:

In Environment 2, six algorithms were tested. The path data for each algorithm are presented in Table 8, while the top view and 3D view of the paths generated by each algorithm are shown in Figs. 13 and 14, respectively.

As shown in Table 8, FPAPF demonstrates relatively stable and efficient path planning performance in Environment 2. It achieves a path length of 210 m and a planning time of 3.06 s, both the shortest and fastest among the compared algorithms. Although its minimum safety distance is slightly lower than those of IAPF, the overall obstacle avoidance process remains smooth, with no high-risk segments close to obstacles. The path safety remains within a reasonable and acceptable range.

From the perspective of spatial path distribution, the visualized results highlight differences in obstacle avoidance strategies among the algorithms. The RRT path exhibits significant fluctuations, with noticeable detours and backtracking. The TAPF path progresses in a relatively direct manner but shows rigidity when navigating dense

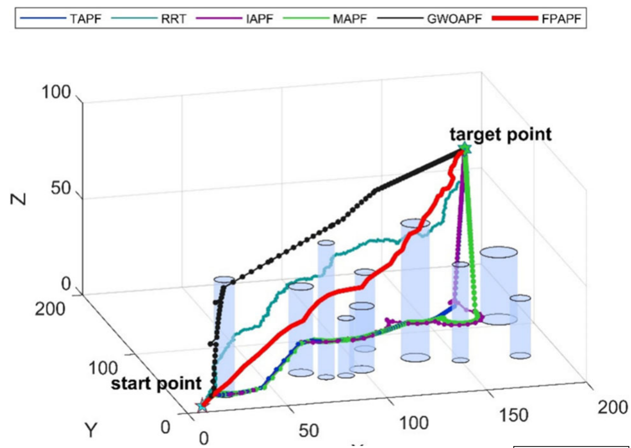


Fig. 14 3D view of paths generated by each algorithm in Environment 2

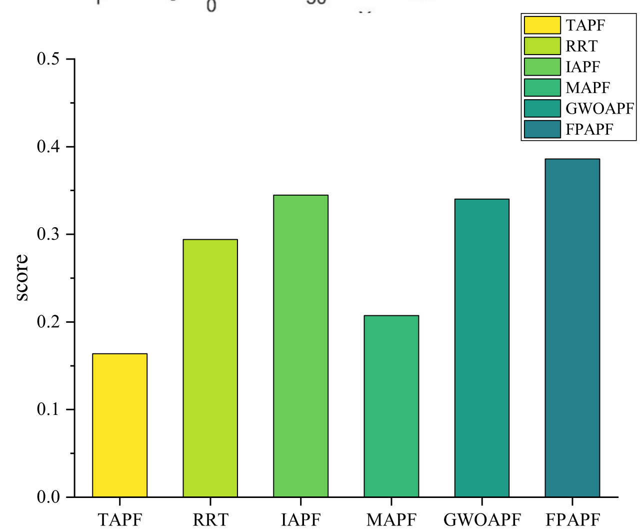


Fig. 15 Weighted scores of algorithms in Environment 2

obstacle areas, resulting in a somewhat inflexible trajectory. IAPF presents extended paths and frequent turning in locally complex regions. In contrast, FPAPF produces a continuous and compact trajectory that smoothly avoids obstacles while maintaining good directional consistency and path clarity.

As shown in Fig. 15, FPAPF achieved the highest overall score, exhibiting the most balanced performance among all algorithms. This indicates that FPAPF effectively coordinates multiple key metrics, including path length, safety distance, and planning time, demonstrating strong overall path planning capability. IAPF ranked second, mainly due to its advantage in safety, particularly its outstanding performance in minimum safety distance. GWOAPF and RRT obtained similar scores. While RRT showed some advantage in average safety distance, its overall score was affected by a smaller minimum distance and insufficient path stability. MAPF delivered stable obstacle avoidance performance but suffered from lower path efficiency, resulting in a moderately low final score. TAPF scored the lowest, primarily due to its longer path length and higher planning time, indicating relatively limited overall performance.

Environment 3 Table 9 presents the coordinate information of each obstacle in Environment 3:

In Environment 3, six algorithms were tested. The path data for each algorithm are presented in Table 10, while the top view and 3D view of the paths generated by each algorithm are shown in Figs. 16 and 17, respectively.

In Environment 3, the TAPF algorithm failed due to local minima in densely obstructed regions, preventing progress toward the goal. IAPF, though improved, also failed as the weakened attractive potential near clustered obstacles hindered convergence. RRT and GWOAPF successfully generated paths but exhibited significant detours, with RRT showing a critically low minimum safety distance of 0.15 m, indicating close-proximity risks. In contrast, FPAPF effectively reached the goal, demonstrating strong robustness against local minima. Its integration of

Table 9 Coordinate information of obstacles in Environment 3

Obstacle no.	Center (*x*, *y*)	Height	Radius
Obstacle 1	(70, 50)	60	6
Obstacle 2	(140, 100)	50	6
Obstacle 3	(45, 35)	60	7
Obstacle 4	(110, 140)	53	6
Obstacle 5	(185, 110)	35	6
Obstacle 6	(100, 95)	20	6
Obstacle 7	(80, 110)	28	6
Obstacle 8	(50, 90)	40	6
Obstacle 9	(100, 115)	90	6
Obstacle 10	(90, 84)	55	8
Obstacle 11	(160, 160)	30	5
Obstacle 12	(150, 180)	69	7

Table 10 Path data of each algorithm in Environment 3

Algorithm	Success	Time (s)	Length (m)	Avg. safety (m)	Min. safety (m)
TAPF	No	–	–	–	–
RRT	Yes	7.94	327	17.48	0.15
IAPF	No	–	–	–	–
MAPF	No	–	–	–	–
GWOAPF	Yes	10.09	411	8.04	0.48
FPAPF	Yes	3.55	276	9.40	1.90

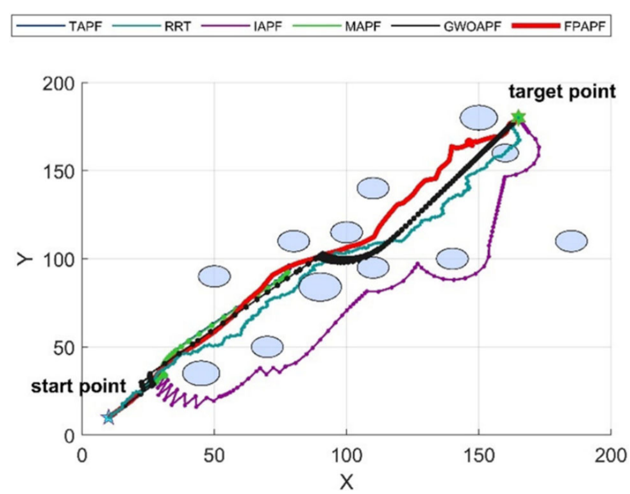


Fig. 16 Top view of paths generated by each algorithm in Environment 3

particle swarm optimization dynamically guides the search direction, helping the algorithm escape trap regions and enhance global convergence.

4.1.2 Experimental Analysis in Dynamic and Hybrid Environments

To evaluate the robustness and obstacle avoidance capability of the proposed path planning algorithm in dynamic environments, multiple dynamic obstacles with varying motion patterns were introduced into a three-dimensional simulation space. All obstacles were modeled as either cylinders or spheres and were dynamically updated during the planning process based on predefined temporal rules.

Fig. 17 3D view of paths generated by each algorithm in Environment 3

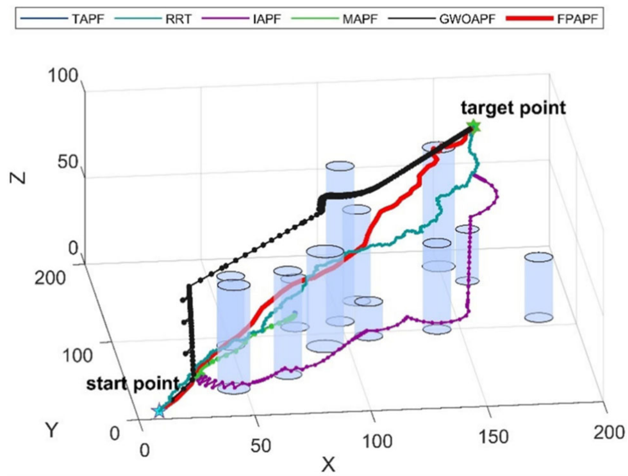


Table 11 Obstacle information in Environment 4

Obstacle no.	Center (x, y)	Height	Radius	Type	Motion description
Obstacle 1	(75, 100)	60	9	Static	–
Obstacle 2	(130, 40)	20	7	Static	–
Obstacle 3	(25, 55)	60	5	Static	–
Obstacle 4	(100, 48)	83	6	Static	–
Obstacle 5	(150, 110)	35	9	Static	–
Obstacle 6	(110, 80)	20	7	Static	–
Obstacle 7	(90, 40)	50	4	Static	–
Obstacle 8	(75, 100)	60	9	Dynamic	Circular motion around (50, 50)
Obstacle 9	(130, 40)	50	7	Dynamic	Linear motion along y-axis
Obstacle 10	(150, 110)	35	9	Dynamic	Circular motion around (100, 100)

Table 12 Path data of each algorithm in Environment 4

Algorithm	Success	Time (s)	Length (m)	Avg. safety (m)	Min. safety (m)
TAPF	Yes	26.12	252	15.31	5.02
RRT	Yes	23.48	269	24.46	1.07
IAPF	Yes	9.79	253	15.47	0.71
MAPF	Yes	20.15	204	22.08	0.17
GWOAPF	Yes	11.28	214	18.47	0.27
FPAPF	Yes	3.89	201	20.72	5.42

A total of three dynamic obstacles and seven static obstacles were deployed. The positions of the dynamic obstacles were computed in real time according to the current simulation time and continuously updated throughout the path planning process, thereby simulating the dynamic characteristics of obstacles in real-world scenarios. Specifically, two obstacles perform circular motion around fixed centers at different angular velocities, while the third obstacle moves back and forth along the y-axis at a constant speed. The remaining obstacles remain stationary. All path planning algorithms were evaluated under the same dynamic environment, with consistent obstacle trajectories, to ensure a fair comparison of their adaptability to dynamic disturbances and obstacle avoidance performance. The initial position of the UAV is set to [10,10,0], and the target point is located at [160,80,90].

Table 11 presents the configuration of obstacles in Environment 4, while Table 12 and Fig. 18 show the algorithm performance metrics and the corresponding trajectory plots under the same scenario.

As shown in Table 10 and Fig. 18, the FPAPF algorithm produces a trajectory with a total length of 201 m in a complex 3D environment containing dynamic obstacles. This value is shorter than those obtained by RRT, MAPF, and IAPF, indicating that the trajectory avoids unnecessary detours. The corresponding planning time is

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Fig. 18 Comparison of paths generated by different algorithms in Environment 4

(a) Top and 3D view of TAPF paths in Environment 4.

(b) Top and 3D view of RRT paths in Environment 4.

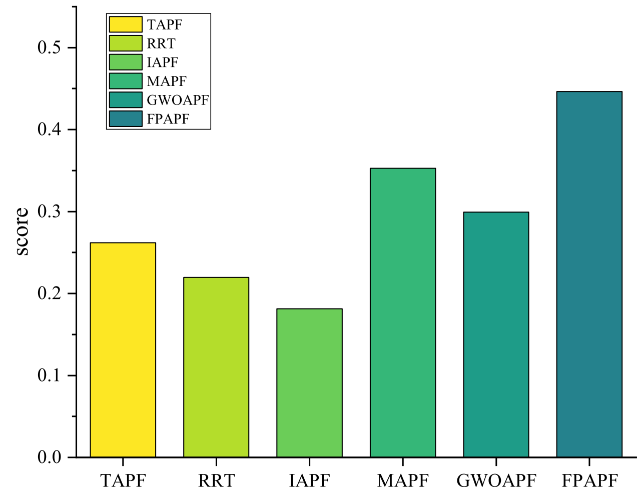
(c) Top and 3D view of IAPF paths in Environment 4.

(d) Top and 3D view of MAPF paths in Environment 4.

(e) Top and 3D view of GWOAPF paths in Environment 4.

(f) Top and 3D view of FPAPF paths in Environment 4.

Fig. 19 Weighted scores of algorithms in Environment 4



3.89 s, which is the lowest among the tested algorithms, reflecting relatively high computational efficiency. In terms of safety, the minimum clearance of the planned trajectory is 5.42 m, which is larger than that of most other approaches.

As illustrated in Fig. 18, the trajectory avoids all static and dynamic obstacles and remains continuous, without oscillations, abrupt turns, or obstacle penetration. These results show that the obtained path is feasible and meets the constraints of the test environment.

To further evaluate the overall performance of each algorithm in dynamic environments, this study adopts the same weighted scoring strategy used in static scenarios, integrating three key metrics: path length, average safety distance, and minimum safety distance. As shown in Fig. 19, FPAPF consistently achieves the highest score, significantly outperforming other benchmark algorithms. This result highlights its superior adaptability and overall stability when confronted with dynamic disturbances. MAPF performs better than in static scenarios, while IAPF shows a noticeable drop in performance. This contrast highlights differences in their effectiveness under dynamic obstacle conditions. TAPF, RRT, and GWOAPF obtain relatively lower scores, primarily due to limited responsiveness to environmental changes.

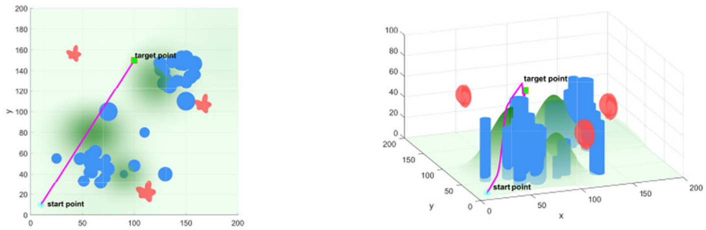
4.1.3 Experimental Analysis in a More Complex Environment

To better approximate real-world scenarios, we construct a more complex setting (hereafter, Environment 5) comprising three components: (i) terrain, formed by the superposition of three Gaussian hills with an added sinusoidal undulation term, yielding a gently varying, differentiable surface; (ii) dense static-obstacle zones, modeled as finite-height right circular cylinders beyond a small base set, two additional high-density clusters are introduced, for a total of 45 static cylinders randomly scattered around the respective cluster centers; and (iii) irregular dynamic obstacles, instantiated as three objects following horizontal elliptical orbits with superimposed vertical sinusoidal motion. The start and target point are set to (10, 10, 0) and (100, 150, 50), respectively.

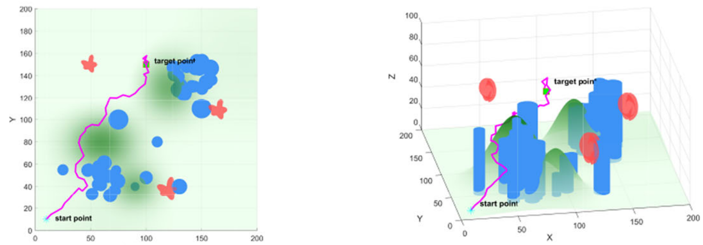
Table 13 and Fig. 20 present the performance metrics and the corresponding trajectory plots for each algorithm in Environment 5, respectively.

As summarized in Table 13, for six algorithms evaluated in a complex 3D environment. FPAPF attains the shortest planning time and comparatively larger clearances, indicating stable obstacle separation without sacrificing feasibility. TAPF and MAPF generate more compact paths yet exhibit very small minimum clearances, implying a higher risk of near-obstacle interactions. RRT and GWOAPF produce longer trajectories with pronounced detours. IAPF achieves higher safety at the cost of additional travel. Overall, FPAPF translates the combination of fastest solve time and superior clearances into greater robustness and operational viability; although its path length is slightly longer than the shortest baselines, it remains the more reliable choice for engineering deployment.

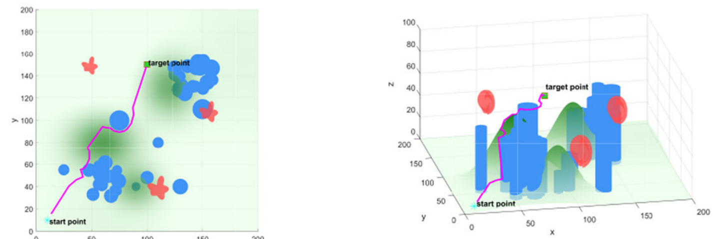
Fig. 20 Comparison of paths generated by different algorithms in Environment 5



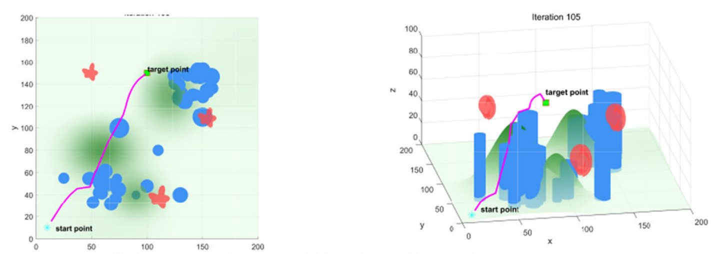
(a) Top and 3D view of TAPF paths in Environment 5.



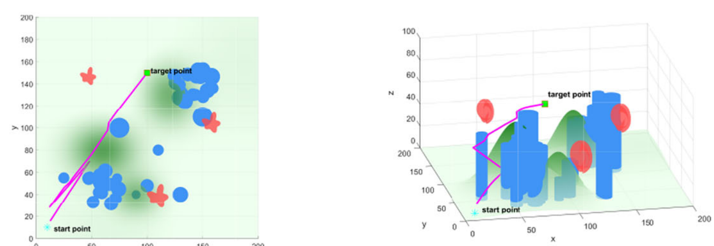
(b) Top and 3D view of RRT paths in Environment 5.



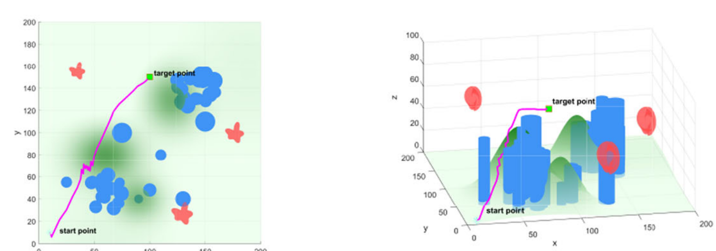
(c) Top and 3D view of IAPF paths in Environment 5.



(d) Top and 3D view of MAPF paths in Environment 5.



(e) Top and 3D view of GWOAPF paths in Environment 5.



(f) Top and 3D view of FPAPF paths in Environment 5.

Table 13 Path data of each algorithm in Environment 5

Algorithm	Success	Time (s)	Length (m)	Avg. safety (m)	Min. safety (m)
TAPF	Yes	65.19	202	6.70	0.10
RRT	Yes	28.77	269	9.42	0.99
IAPF	Yes	28.08	221	17.65	0.20
MAPF	Yes	29.68	204	18.28	0.01
GWOAPF	Yes	25.23	282	17.87	0.05
FPAPF	Yes	13.67	224	16.54	4.45

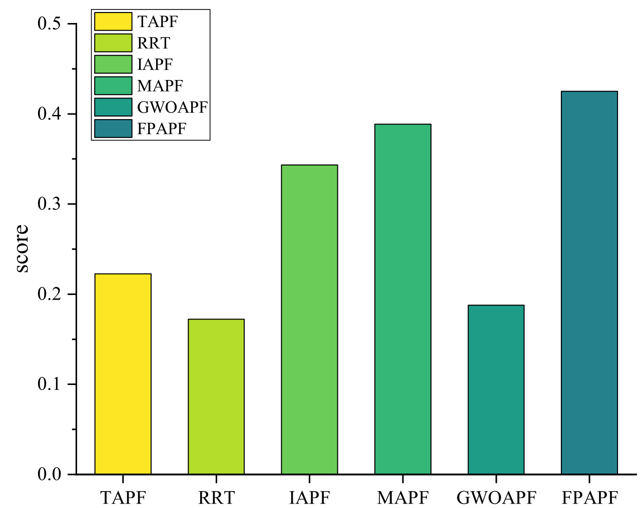


Fig. 21 Weighted scores of algorithms in Environment 5

As shown in the Fig. 21, FPAPF again attains the highest weighted score, indicating that its path length remains sufficiently compact while safety metrics contribute substantially, thereby leading the weighted aggregate. MAPF and IAPF rank next, benefiting from either shorter paths or larger average clearances. TAPF is penalized for small minimum clearances, whereas RRT and GWOAPF obtain lower overall scores due to longer trajectories. Accordingly, under a weighting scheme that assigns greater importance to path length, FPAPF still achieves the best overall performance. These results suggest that, in complex and dynamic 3D environments, FPAPF combines strong geometric optimization with stable safety margins to balance efficiency and reliability, making it well suited to real-time, risk-sensitive flight missions.

In summary, the FPAPF algorithm demonstrates excellent performance across static, dynamic, and more complex environments. It consistently achieves the lowest computational time, reflecting high planning efficiency. In terms of path length, FPAPF maintains the shortest routes in static and dynamic environments, with only a slight increase in more complex settings. Regarding safety distance, it shows relatively favorable performance in static environments and ranks among the best in dynamic and complex scenarios, confirming its reliable flight safety. Overall, FPAPF exhibits strong adaptability across different environments and represents a promising solution for path planning in complex scenarios.

4.2 Analysis of Path Executability and Controller Performance

4.2.1 Experimental Setup

To evaluate the effectiveness of the proposed path executability assessment framework, comparative tracking experiments were conducted using three controllers: PID, LQR, and ELQR. The test trajectory was generated by the FPAPF algorithm in Environment 1 and retained its original nonsmooth characteristics to simulate realistic path

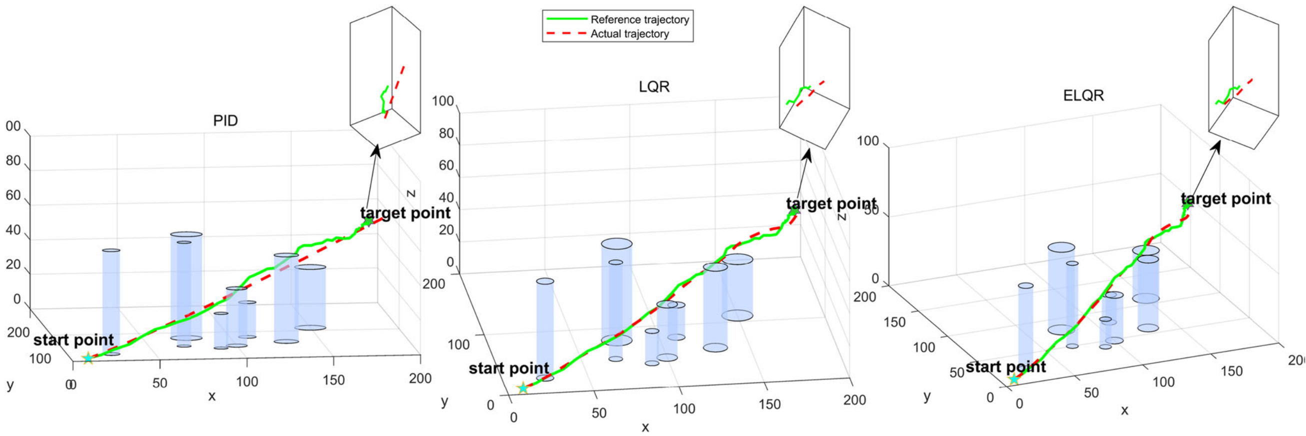
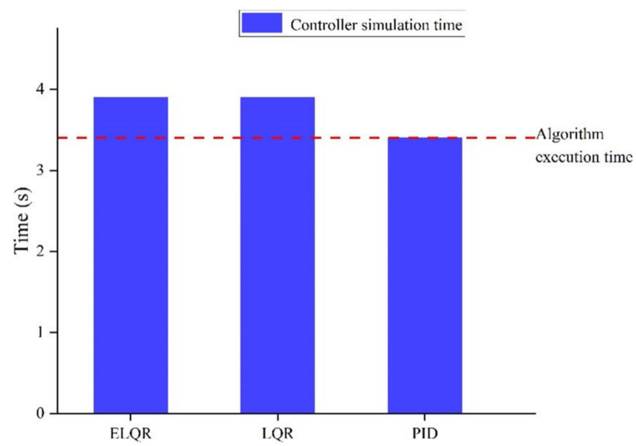


Fig. 22 3D trajectory tracking results of different controllers

Fig. 23 Comparison between controller simulation time and path generation time



conditions. To further replicate the challenges of complex environments, observation noise and mass disturbances were introduced during the simulation process.

4.2.2 Path Executability Verification

To verify the executability of the proposed path under different control strategies, this study evaluates the trajectory from two perspectives: trackability and simulated execution time.

Figure 22 presents the 3D trajectory tracking results of the PID, LQR, and ELQR controllers under a unified path input. It can be observed that all three controllers successfully guide the UAV from the starting point to the destination while avoiding all obstacles, indicating that the proposed path demonstrates good dynamic executability under disturbances and uncertainties. Notably, the ELQR controller maintains accurate trajectory tracking even in non-smooth segments, with the actual trajectory closely aligning with the reference. This highlights the enhanced predictive compensation strategy's superior robustness and responsiveness when dealing with sharp turns and dynamic disturbances.

Figure 23 verifies the path executability from a temporal perspective. The red dashed line indicates the time consumed by the algorithm to generate the path, while the blue bars represent the simulated execution times required by each controller to complete the trajectory. As shown, the PID controller achieves the shortest simulation time at 3.4 s, closely matching the path generation time. Although the LQR and ELQR controllers consume slightly more time, their execution remains within a reasonable range, with no failure to complete the path within the time constraint. This confirms that the trajectory is temporally feasible across different control strategies and satisfies real-time requirements for practical deployment.

Table 14 Comparative statistics of three controllers

	RMSE	Max_err	Final_err	Mean_jerk	Max_jerk	Robustness	DAR
PID	62.494	107.91	107.91	35.455	65.462	0.0011067	7.7972e−5
LQR	6.1948	10.47	5.5022	66.539	268.7	0.19005	0.0048055
ELQR	5.4567	8.1312	5.1389	64.612	262.53	0.24359	0.0080247

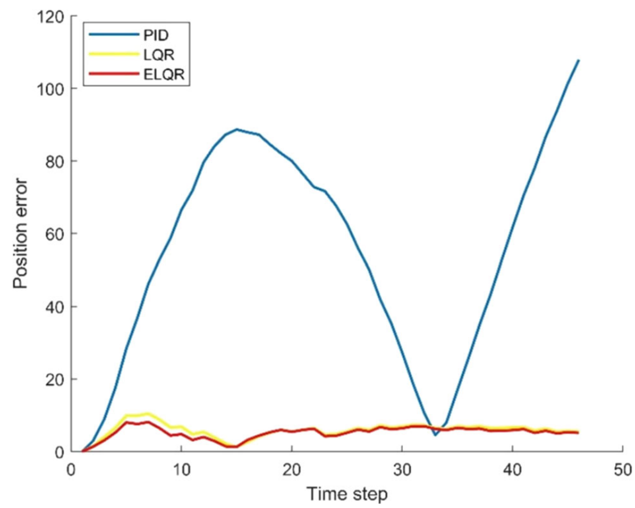


Fig. 24 Comparison of position tracking error under different controllers

In summary, the proposed path demonstrates excellent executability, allowing multiple controllers to stably and promptly track the trajectory under perturbed conditions. This provides a reliable foundation for subsequent performance comparisons of control strategies.

4.2.3 Analysis of Controller Tracking Results

To comprehensively evaluate the control performance, the experiment collected all relevant metrics described in Sect. 3.5.4 to characterize the trackability of the proposed path under different control strategies. The Table 14 summarizes the statistical results of various indicators for the three controllers.

The experimental results indicate that the traditional PID controller performs poorly in terms of average error, maximum error, and final error, demonstrating its limited ability to cope with disturbances and abrupt trajectory changes in complex environments. In contrast, the LQR controller significantly improves tracking accuracy and achieves notable enhancements in both robustness and disturbance attenuation ratio (DAR). Furthermore, the ELQR controller proposed in this study introduces a velocity feedforward compensation mechanism, enabling the system to respond more proactively to upcoming trajectory variations. As a result, it further optimizes all error-related metrics and achieves the best performance in terms of robustness and DAR, demonstrating superior disturbance rejection capability and control stability. Although the LQR-based controllers exhibit larger variations in acceleration due to their pursuit of high-precision tracking, the overall smoothness remains within acceptable limits. These findings validate the feasibility and advantages of using such controllers as a reference baseline for path executability evaluation in complex environments.

Figure 24 illustrates the real-time position error comparison of different controllers when tracking the same path. The traditional PID controller exhibits rapid error accumulation in the early stage and significant deviations in the later stages, with a maximum error exceeding 100. This reflects its lack of robustness when facing non-smooth trajectories and environmental disturbances. In contrast, the standard LQR controller effectively suppresses error growth, though slight fluctuations are still observed in certain segments. The ELQR controller maintains a

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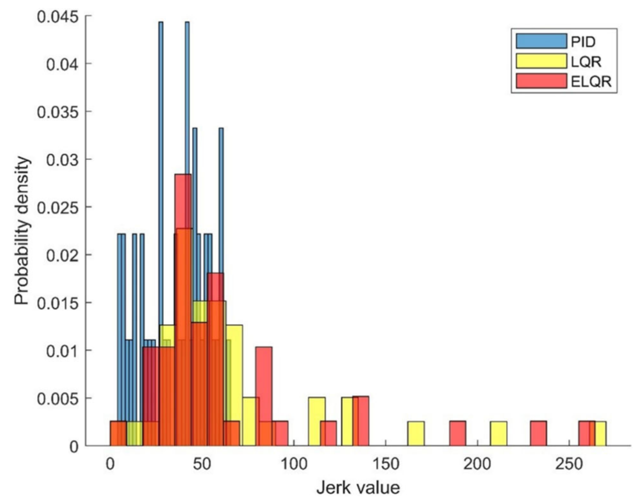

Fig. 25 Histogram of Jerk values for different controllers

consistently low and stable error curve throughout the trajectory, indicating its stronger feedforward adaptability and disturbance rejection capability, which enables stable and high-precision tracking performance.

Figure 25 illustrates the probability density distribution of jerk generated by the three controllers during the path tracking process, aiming to evaluate the smoothness performance of each control strategy during dynamic response. The results show that while the traditional PID controller exhibits a high frequency in the low-jerk region, its distribution is sharply peaked, indicating pronounced oscillations in the control process that may hinder system stability. In contrast, the standard LQR controller presents a broader distribution in higher jerk regions, demonstrating strong regulation capabilities but at the cost of larger control effort to minimize tracking error. The ELQR controller exhibits superior smoothness, with a concentrated jerk distribution peak and rapidly decaying tails, indicating effective suppression of transient shocks while maintaining fast response. This reflects its enhanced control compliance and adaptability to disturbances. The jerk distribution results further validate the advantages of the enhanced control strategy in complex path tracking tasks.

Through the comparative experiments of the three controllers, the executability of the non-smooth path under complex disturbance conditions is validated. The results demonstrate that all controllers successfully accomplish the path tracking task, indicating that the path possesses good executability. Among them, the ELQR controller achieves the best performance in error control, smoothness, and disturbance attenuation, highlighting its superior robustness and strong potential for practical applications.

4.2.4 Comparative Evaluation of Algorithms Under the ELQR Controller

To compare the trajectory tracking performance of six algorithms under identical disturbance conditions, the evaluation metrics defined in Sect. 3.5.4 were employed. Table 15 presents the summary statistics of all metrics under ELQR controller. Figure 26 illustrates the histograms of four key indicators, where the darker bars denote the algorithm achieving the best performance for each respective metric.

The comparative results indicate that FPAPF and GWOAPF deliver the strongest overall practical performance, albeit with different emphases. In terms of tracking accuracy, GWOAPF is marginally superior to FPAPF; however, FPAPF exhibits markedly better peak-error and motion-smoothness characteristics (lower jerk) and achieves the highest robustness score with the smallest error variability. For disturbance rejection, GWOAPF ranks first, and RRT is also competitive, though at the cost of larger jerk and less smooth execution. Overall, when overall accuracy and disturbance rejection are prioritized, GWOAPF is preferable; when engineering considerations demand reduced oscillation and lower mechanical shock-thereby easing control effort and improving operational executability-FPAPF is the better choice.

Table 15 Algorithm comparison under the ELQR controller

	RMSE	Max_err	Final_err	Mean_jerk	Max_jerk	Robustness	DAR
TAPF	1.0831	2.2469	1.0357	60.58	772.66	0.85586	0.22963
RRT	0.95085	1.6546	1.2350	319.01	760.54	0.94271	0.29867
IAPF	1.0759	2.8974	0.97344	62.079	807.26	0.85925	0.23329
MAPF	1.0976	2.3659	0.94027	66.068	876.04	0.84153	0.22389
GWOAPF	0.94702	2.8165	0.86640	121.52	901.95	0.88865	0.30020
FPAPF	0.97824	1.3603	1.07290	51.75	447.35	0.94866	0.27688

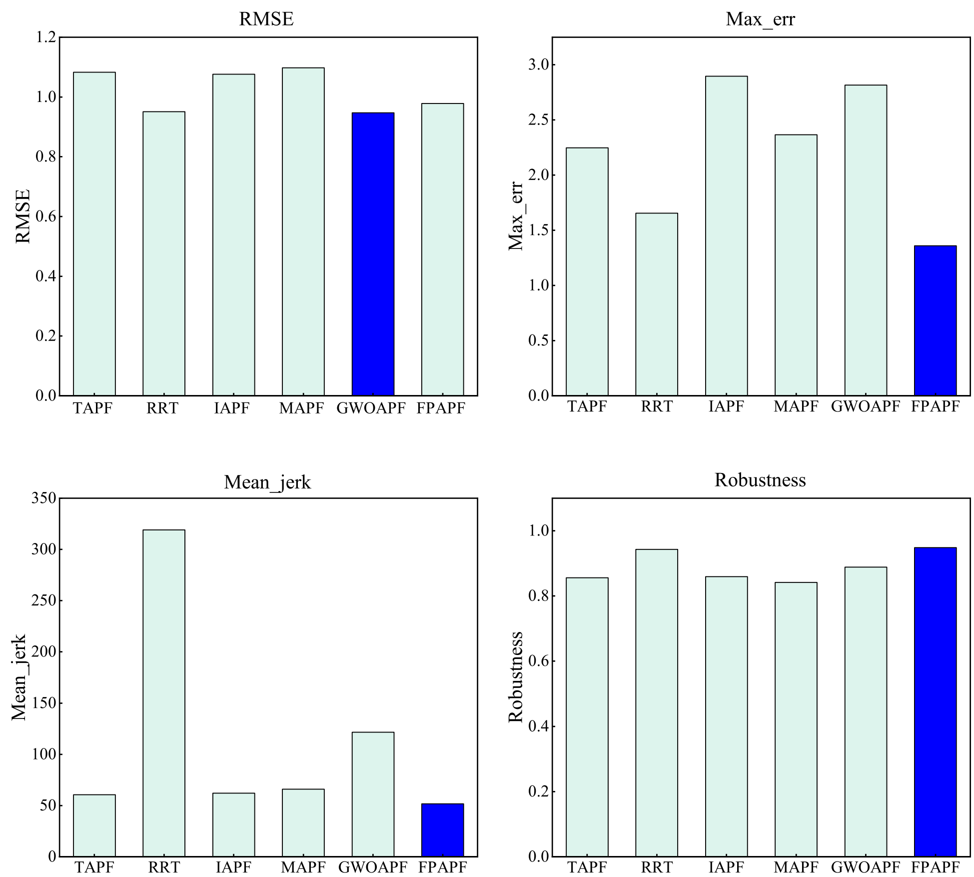

Fig. 26 Histograms of the four key performance indicators under the ELQR controllor

5 Conclusion

This study proposes an improved path generation method that integrates APF with PSO to address UAV path planning challenges in complex three-dimensional environments. The introduction of a Gaussian repulsive potential enhances the continuity of the potential field near obstacle boundaries, thereby improving the stability and flexibility of obstacle avoidance. In addition, the PSO-based search mechanism effectively alleviates the limitations of traditional APF, such as susceptibility to local minima and target inaccessibility. A multi-criteria evaluation framework was developed to assess path efficiency and safety, incorporating metrics such as path length, average safety distance, and minimum safety distance. Simulation results validated the advantages of the proposed method in feasibility, safety, and global planning capability across different scenarios.

Furthermore, controller-level tracking experiments with PID, LQR, and ELQR further verified the executability of planned paths under disturbances and non-smooth trajectories. However, limitations remain: in highly complex environments, path length may increase compared with other algorithms, and computational burdens become more significant in dynamic settings. Overall, the proposed method demonstrates strong adaptability and practical feasibility, while the identified limitations highlight avenues for further improvement. Future work will focus on enhancing adaptability through learning-based parameter tuning, reducing computational overhead via parallel and distributed optimization, and extending the framework to multi-UAV coordination and energy-aware path planning.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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